ECE330 EXAM 2 EQUATION SHEET

sin(x)=cos(x-90°) $\quad \bar{V} = \bar{Z}\bar{I} \quad \bar{S} = \bar{V}\bar{I}^* = P + jQ \quad \bar{S}_{3\varphi} = \sqrt{3}V_L I_L \angle\theta$

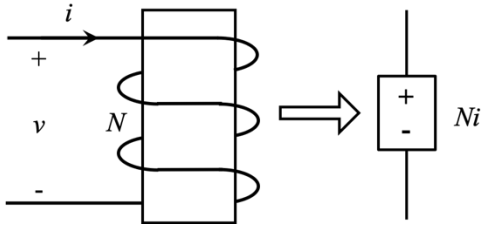
$0<\theta<180°$ (lag) $\quad I_L = \sqrt{3}I_\varphi$ (delta) $\quad \bar{Z}_Y = \bar{Z}_\Delta/3$

$-180°<\theta<0$ (lead) $\quad V_L = \sqrt{3}V_\varphi$ (wye) $\quad \mu_0 = 4\pi \times 10^{-7}$H/m

ABC phase sequence has A at 0, B at -120°, and C at +120°

$\mathcal{R} = \frac{l}{\mu A} \quad Ni = \mathcal{R}\phi \quad \phi = BA$

$\lambda = N\phi = Li$ (if linear) $\quad v = \frac{d\lambda}{dt} \quad k = \frac{M}{\sqrt{L_1 L_2}}$



$W_m = \int_0^\lambda i d\hat{\lambda}, \quad W_m' = \int_0^i \lambda d\hat{\imath}, \quad i = \frac{\partial W_m}{\partial \lambda}, \quad \lambda = \frac{\partial W_m'}{\partial i}, \quad W_m + W_m' = i\lambda.$

If linear, $W_m = W_m' = \frac{1}{2}Li^2 = \frac{1}{2}L^{-1}\lambda^2 = \frac{1}{2}\lambda i.$

$f^e = -\frac{\partial W_m}{\partial x} = +\frac{\partial W_m'}{\partial x}, \quad \tau^e = -\frac{\partial W_m}{\partial \theta} = +\frac{\partial W_m'}{\partial \theta}.$

$W_m(\lambda_1, \lambda_2, x) = \int_0^{\lambda_1} i_1(\hat{\lambda}_1, 0, x) d\hat{\lambda}_1 + \int_0^{\lambda_2} i_2(\lambda_1, \hat{\lambda}_2, x) d\hat{\lambda}_2,$

$W_m'(i_1, i_2, x) = \int_0^{i_1} \lambda_1(\hat{\imath}_1, 0, x) d\hat{\imath}_1 + \int_0^{i_2} \lambda_2(i_1, \hat{\imath}_2, x) d\hat{\imath}_2,$

$W_m + W_m' = i_1\lambda_1 + i_2\lambda_2.$

$EFE_{a\to b} = \int_{a\to b} i d\lambda, \quad EFM_{a\to b} = \int_{a\to b} -f^e dx, \quad EFE_{a\to b} + EFM_{a\to b} = W_{mb} - W_{ma}.$